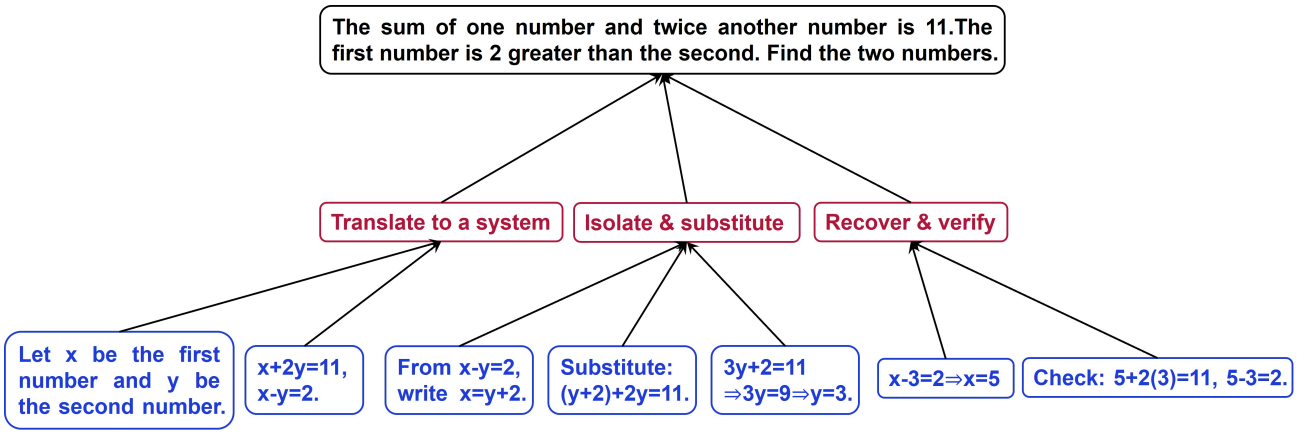

Figure H6. **A concrete algebra example illustrating skills, routing mechanisms, and identification conditions.** The **black box** denotes the problem descriptor $\mathbf{P}$. The **red boxes** denote latent selection variables. The **blue boxes** denote observed trace tokens. Here we use some examples to explain three identification conditions. **Condition-Neighborhood coverage**: If *"Translate to a system"* only ever appears right before *"Isolate & substitute"*, it can be mistaken as one fused routine; seeing *"Translate to a system"* also before *"Eliminate by subtraction"* ($(x+2y)-(x-y)=11-2 \Rightarrow 3y=9 \Rightarrow y=3$) provides distinct neighborhoods that isolate *"Translate to a system"* as a reusable module. **Condition-Local distinguishability**: For each selection variable in the figure, at least one lower-level module is conditioned only on it; for example, *"Substitute:* $(y+2)+2y=11$*"* is only conditioned on *"Isolate & substitute"*. **Condition-Restricted choice sets**: At each intermediate step, only a limited set of plausible next modules or trace continuations are available algebraically. For instance, upon completing the step *"Substitute:* $(y+2)+2y=11$*"*, we naturally and logically proceed to the step *"*$3y+2=11 \Rightarrow 3y=9 \Rightarrow y=3$*"*.

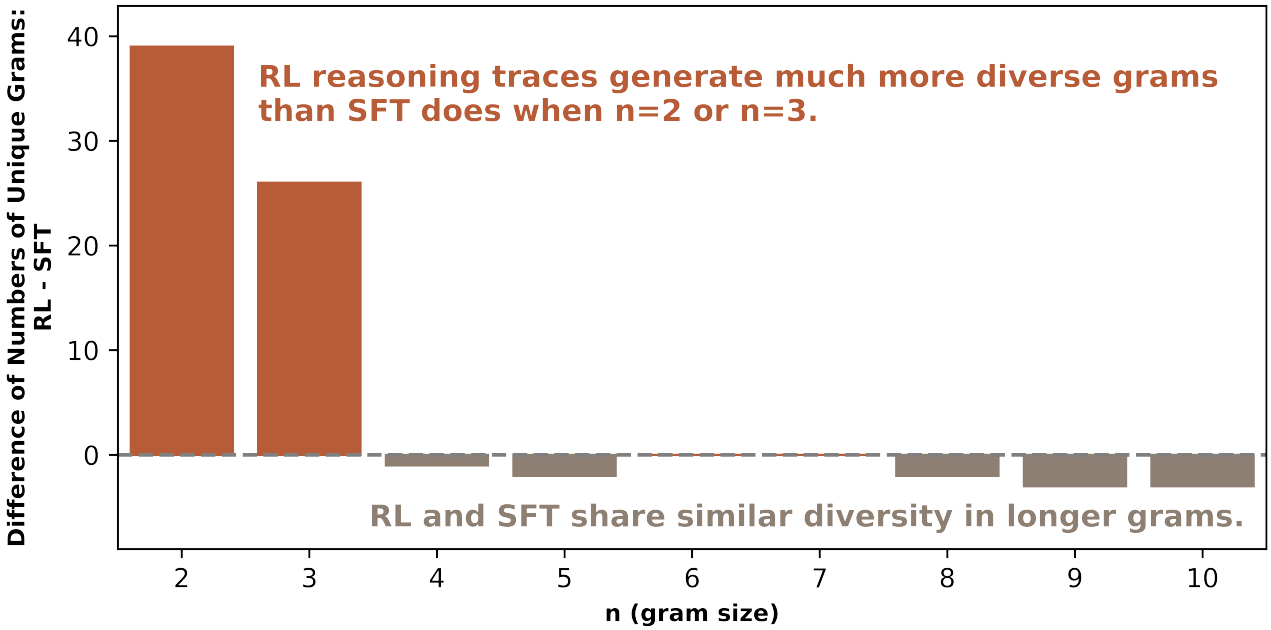

Figure H7. **Difference of numbers of unique grams between the RL and SFT model.** For the n-grams obtained in Appendix D (lines 979-983), we calculate the number of unique grams for both RL and SFT models, then visualize their differences here. The results show that (1) RL reasoning traces generate much more diverse grams than SFT does when $n=2$ and $n=3$; (2) RL and SFT reasoning traces share similar diversity in longer grams.